ASE CHAPTER 7

T7 Software Reliability Calculations Pack

Problem 7.1 · Basic Execution Time Model — Current Failure Intensity Evaluation

Assume that the initial failure intensity is $\lambda_0 = 10\ \text{failures/CPU hr}$ and the total number of failures expected over infinite time is $v_0 = 100\ \text{failures}$.

1. Calculate the failure intensity after 10 failures have been experienced.

2. Calculate the failure intensity after 50 failures have been experienced.

1. **Given:**
   - Initial failure intensity: $\lambda_0 = 10\ \text{failures/CPU hr}$
   - Total expected failures: $v_0 = 100\ \text{failures}$
   - Failure counts experienced: $\mu_1 = 10\ \text{failures}$, $\mu_2 = 50\ \text{failures}$
2. **Required:**
   - Failure intensity $\lambda(10)$ in failures/CPU hr.
   - Failure intensity $\lambda(50)$ in failures/CPU hr.
3. **Model & Assumptions:**
   - **Musa Basic Execution Time Model.**
   - The model assumes a finite expected failure count and a linear reduction in failure intensity as failures are experienced.
4. **Equation:**

$$\lambda(\mu) = \lambda_0 \left(1 - \frac{\mu}{v_0}\right)$$

5. **Substitution & Working:**
   - For $\mu_1 = 10\ \text{failures}$:

$$\lambda(10) = 10 \times \left(1 - \frac{10}{100}\right) = 10 \times (1 - 0.10) = 10 \times 0.90 = 9.0\ \text{failures/CPU hr}$$

   - For $\mu_2 = 50\ \text{failures}$:

$$\lambda(50) = 10 \times \left(1 - \frac{50}{100}\right) = 10 \times (1 - 0.50) = 10 \times 0.50 = 5.0\ \text{failures/CPU hr}$$

6. **Result:**
   - $\lambda(10) = 9.0\ \text{failures/CPU hr}$
   - $\lambda(50) = 5.0\ \text{failures/CPU hr}$
7. **Check / Interpretation:** When $\mu = 0$, $\lambda = 10\ \text{failures/CPU hr}$ (λ_0). When half the faults are removed ($\mu = 50 = v_0/2$), the failure intensity drops by exactly half to $5.0\ \text{failures/CPU hr}$, verifying the strictly linear property of the Basic Execution Time Model.

Source: Aggarwal & Singh (3rd ed.), §7.3.1, Example 7.1, p. 334; companion slides 47–54.

Problem 7.2 · Failure Intensity from Decay Parameter (Exercise 7.16)

Assume that the initial failure intensity is $\lambda_0 = 6\ \text{failures/CPU hr}$. The failure intensity decay parameter is $\theta = 0.02/\text{failure}$. We assume that 45 failures have been experienced. Calculate the current failure intensity.

1. **Given:**
   - Initial failure intensity: $\lambda_0 = 6\ \text{failures/CPU hr}$

- Failure intensity decay parameter: $\theta = 0.02/\text{failure}$
- Failures experienced: $\mu = 45\ \text{failures}$

2. **Required:**
   - Current failure intensity $\lambda(45)$ in failures/CPU hr.
3. **Model & Assumptions:**
   - **Logarithmic Poisson Execution Time Model (Musa–Okumoto).**
     - The decay parameter θ defines the exponential reduction in failure intensity per failure experienced.
4. **Equation (Logarithmic Poisson Model):**

$$\lambda(\mu) = \lambda_0 \exp(-\theta\mu)$$

5. **Substitution & Working:**

$$\lambda(45) = 6 \times \exp(-0.02 \times 45) = 6 \times \exp(-0.90)$$

Since $\exp(-0.90) \approx 0.40656966$:

$$\lambda(45) = 6 \times 0.40656966 \approx 2.4394\ \text{failures/CPU hr}$$

6. **Result:**
   - $\lambda(45) \approx 2.44\ \text{failures/CPU hr}$.
7. **Check / Interpretation:**
   - Because earlier failures have a greater impact on reducing failure intensity in the logarithmic model than later ones, 45 failures reduce intensity from 6 down to ≈ 2.44, which represents a $\approx 59.3\%$ reduction.

Source: Aggarwal & Singh (3rd ed.), Exercise 7.16, p. 363; companion slide 122.

Problem 7.3 · Multi-Stage Basic Execution Time Evaluation (Exercise 7.18)

Assume that a program will experience 150 failures in infinite time. It has now experienced 80 failures. The initial failure intensity was 10 failures/CPU hr.

1. Determine the current failure intensity.

2. Calculate the failures experienced and failure intensity after 25 and 40 CPU hrs of execution from time zero.

3. Compute additional failures and additional execution time required to reach the failure intensity objective of 2 failures/CPU hr from the current state.

1. **Given:**
   - Total failures over infinite time: $v_0 = 150\ \text{failures}$
   - Initial failure intensity: $\lambda_0 = 10\ \text{failures/CPU hr}$
   - Failures experienced currently: $\mu_P = 80\ \text{failures}$
   - Evaluation execution times: $\tau_1 = 25\ \text{CPU hrs}$, $\tau_2 = 40\ \text{CPU hrs}$
   - Target failure intensity objective: $\lambda_F = 2\ \text{failures/CPU hr}$
2. **Required:**
   - (i) Current failure intensity λ_P.
   - (ii) Failures experienced $\mu(25)$, $\mu(40)$ and failure intensity $\lambda(25)$, $\lambda(40)$.
   - (iii) Additional failures $\Delta\mu$ and additional execution time $\Delta\tau$ to reach $\lambda_F = 2\ \text{failures/CPU hr}$.
3. **Model & Assumptions:**
   - **Musa Basic Execution Time Model.**
   - All relations express failures experienced μ and failure intensity λ as functions of execution time τ.
4. **Equations:**
   - Current intensity from experienced failures:

$$\lambda(\mu) = \lambda_0 \left(1 - \frac{\mu}{v_0}\right)$$

- Intensity and failures as functions of execution time τ:

$$\lambda(\tau) = \lambda_0 \exp\left(-\frac{\lambda_0}{v_0}\tau\right)$$

$$\mu(\tau) = v_0 \left[1 - \exp\left(-\frac{\lambda_0}{v_0}\tau\right)\right]$$

- Increments between present state (λ_P) and target objective (λ_F):

$$\Delta\mu = \frac{v_0}{\lambda_0}(\lambda_P - \lambda_F)$$

$$\Delta\tau = \frac{v_0}{\lambda_0} \ln\left(\frac{\lambda_P}{\lambda_F}\right)$$

5. **Substitution & Working:**

- **Part (i): Current failure intensity:**

$$\lambda_P = 10 \times \left(1 - \frac{80}{150}\right) = 10 \times \left(1 - \frac{8}{15}\right) = 10 \times \frac{7}{15} = \frac{70}{15} \approx 4.6667 \approx 4.67 \text{ failures/CPU hr}$$

- **Part (ii): Values at $\tau = 25$ and $\tau = 40$ CPU hrs:** Ratio $\frac{\lambda_0}{v_0} = \frac{10}{150} = \frac{1}{15} \approx 0.066667 \text{ hr}^{-1}$.
  - At $\tau = 25$ CPU hrs:

$$\frac{\lambda_0}{v_0}\tau = \frac{25}{15} = \frac{5}{3} \approx 1.66667$$

$$\lambda(25) = 10 \times \exp\left(-\frac{5}{3}\right) \approx 10 \times 0.188876 = 1.8888 \approx 1.89 \text{ failures/CPU hr}$$

$$\mu(25) = 150 \times \left[1 - \exp\left(-\frac{5}{3}\right)\right] = 150 \times (1 - 0.188876) = 150 \times 0.811124 = 121.669 \approx 121.67 \text{ failures}$$

  - At $\tau = 40$ CPU hrs:

$$\frac{\lambda_0}{v_0}\tau = \frac{40}{15} = \frac{8}{3} \approx 2.66667$$

$$\lambda(40) = 10 \times \exp\left(-\frac{8}{3}\right) \approx 10 \times 0.069483 = 0.6948 \approx 0.69 \text{ failures/CPU hr}$$

$$\mu(40) = 150 \times \left[1 - \exp\left(-\frac{8}{3}\right)\right] = 150 \times (1 - 0.069483) = 150 \times 0.930517 = 139.578 \approx 139.58 \text{ failures}$$

- **Part (iii): Additional failures and execution time to reach $\lambda_F = 2$ failures/CPU hr:**
  - Using current state $\lambda_P = \frac{70}{15} \approx 4.6667$ failures/CPU hr and $\mu_P = 80$:

$$\Delta\mu = \frac{150}{10} \times \left(\frac{70}{15} - 2\right) = 15 \times \left(\frac{40}{15}\right) = 40 \text{ failures}$$

  (Direct cross-check: Target failures $\mu_F = 150 \times (1 - 2/10) = 150 \times 0.8 = 120$; $\Delta\mu = 120 - 80 = 40$ failures).
  - **Additional execution time:**
    $\Delta\tau = \frac{150}{10} \times \ln\left(\frac{4.6667}{2}\right) = 15 \times \ln(2.33333) = 15 \times 0.847298 \approx 12.7095 \approx 12.71$ CPU hrs

6. **Result:**

- (i) $\lambda_P \approx 4.67$ failures/CPU hr
- (ii) At $\tau = 25$: $\lambda(25) \approx 1.89$ failures/CPU hr, $\mu(25) \approx 121.67$ failures;
  At $\tau = 40$: $\lambda(40) \approx 0.69$ failures/CPU hr, $\mu(40) \approx 139.58$ failures.
- (iii) $\Delta\mu = 40$ failures, $\Delta\tau \approx 12.71$ CPU hrs.

7. Check / Interpretation: Target intensity $2\ \text{failures/CPU hr}$ is reached at total execution time $\tau_F = 15\ln(10/2) \approx 24.14$ CPU hours. The current state ($\mu_P = 80$, $\lambda_P = 70/15$) occurs at $\tau_P = 15\ln(10/(70/15)) \approx 11.43$ CPU hours, so the additional time is $\Delta\tau = \tau_F - \tau_P \approx 12.71$ CPU hours, consistent with the result above.

Source: Aggarwal & Singh (3rd ed.), Exercise 7.18, p. 363; companion slide 123.

Problem 7.4 · Release Objective Evaluation with Unit Conversion (Exercise 7.24)

A program is expected to have 250 faults. It is also assumed that one fault may lead to one failure. The initial failure intensity is 5 failures/CPU hr. The program is released with a failure intensity objective of 4 failures per 10 CPU hours. Calculate the number of failures experienced before release.

1. **Given:**
   - Total faults / failures over infinite time: $v_0 = 250$ failures (1 fault = 1 failure assumption).
   - Initial failure intensity: $\lambda_0 = 5\ \text{failures/CPU hr}$.
   - Release objective: 4 failures per 10 CPU hours.
2. **Required:**
   - Number of failures experienced before release (μ_F).
3. **Model & Assumptions:**
   - **Musa Basic Execution Time Model.**
   - Units must be strictly normalized to failures/CPU hr before substitution.
4. **Unit Normalization & Equation:**

$$\lambda_F = \frac{4\ \text{failures}}{10\ \text{CPU hrs}} = 0.40\ \text{failures/CPU hr}$$

$$\lambda(\mu) = \lambda_0\left(1 - \frac{\mu}{v_0}\right) \implies \mu = v_0\left(1 - \frac{\lambda_F}{\lambda_0}\right)$$

5. **Substitution & Working:**

$$\mu_F = 250 \times \left(1 - \frac{0.40}{5.0}\right) = 250 \times (1 - 0.08) = 250 \times 0.92 = 230\ \text{failures}$$

6. **Result:**
   - $\mu_F = 230\ \text{failures experienced before release.}$
7. **Check / Interpretation:**
   - The remaining unmanifested faults at release are $v_0 - \mu_F = 250 - 230 = 20\ \text{faults}$.
   - The fraction of faults removed is $230/250 = 92\%$, which corresponds to an intensity reduction from 5.0 to 0.40 (92% reduction).

Source: Aggarwal & Singh (3rd ed.), Exercise 7.24, p. 364; companion slide 125.

Problem 7.5 · Logarithmic Poisson Model — Failures Experienced after 20 CPU Hours

Assume that the initial failure intensity is $\lambda_0 = 20\ \text{failures/CPU hr}$ and the failure intensity decay parameter is $\theta = 0.02/\text{failure}$. Determine the number of failures experienced after 20 CPU hours of execution.

1. **Given:**
   - Initial failure intensity: $\lambda_0 = 20\ \text{failures/CPU hr}$
   - Failure intensity decay parameter: $\theta = 0.02/\text{failure}$
   - Execution time: $\tau = 20\ \text{CPU hrs}$
2. **Required:**
   - Expected number of failures experienced $\mu(20)$.
3. **Model & Assumptions:**

- **Musa–Okumoto Logarithmic Poisson Execution Time Model.**
- The failure intensity decreases exponentially with failures experienced.

4. **Equation:**

$$\mu(\tau) = \frac{1}{\theta}\ln(\lambda_0\theta\tau + 1)$$

5. **Substitution & Working:**
   - Inner dimensionless term:

$$\lambda_0\theta\tau + 1 = 20 \times 0.02 \times 20 + 1 = 8 + 1 = 9$$

   - Logarithm evaluation:

$$\ln(9) \approx 2.19722458$$

   - Multiplication:

$$\mu(20) = \frac{1}{0.02} \times 2.19722458 = 50 \times 2.19722458 = 109.8612 \approx 109.86 \text{ failures}$$

6. **Result:**
   - $\mu(20) = 109.86 \text{ failures} \approx 110 \text{ failures}$.
   - The displayed worked integer is 109; direct substitution into the stated equation gives 109.86.
7. **Check / Interpretation:**
   - The result follows from $\lambda_0\theta\tau + 1 = 9$ and is consistent with the logarithmic-model equation.

Source: Aggarwal & Singh (3rd ed.), §7.3.2, Example 7.2, pp. 336–337; companion slide 62.

Problem 7.6 · Logarithmic Poisson Model — Failure Intensity at Specific Execution Time

Using the parameters of Problem 7.5 ($\lambda_0 = 20 \text{ failures/CPU hr}$, $\theta = 0.02/\text{failure}$), calculate the failure intensity after 20 CPU hours of execution.

1. **Given:**
   - Initial failure intensity: $\lambda_0 = 20 \text{ failures/CPU hr}$
   - Failure intensity decay parameter: $\theta = 0.02/\text{failure}$
   - Execution time: $\tau = 20 \text{ CPU hrs}$
2. **Required:**
   - Failure intensity $\lambda(20)$ in failures/CPU hr.
3. **Model & Assumptions:**
   - **Musa–Okumoto Logarithmic Poisson Execution Time Model.**
4. **Equations:**
   - Method 1 (Direct function of execution time τ):

$$\lambda(\tau) = \frac{\lambda_0}{\lambda_0\theta\tau + 1}$$

   - Method 2 (Function of failures experienced μ):

$$\lambda(\mu) = \lambda_0 \exp(-\theta\mu)$$

5. **Substitution & Working:**
   - **Method 1 (Direct from $\tau = 20$):**

$$\lambda(20) = \frac{20}{20 \times 0.02 \times 20 + 1} = \frac{20}{9} \approx 2.2222 \text{ failures/CPU hr}$$

- **Method 2 (using the unrounded $\mu(20) = 109.8612$ from Problem 7.5):**

$$\lambda(20) = 20\exp(-0.02 \times 109.8612) \approx 2.2222 \text{ failures/CPU hr}$$

6. **Result:**
   - $\lambda(20) \approx 2.22 \text{ failures/CPU hr}$.
7. **Check / Interpretation:**
   - The two equivalent forms agree: $\lambda(20) = \lambda_0/(\lambda_0\theta\tau + 1) = \lambda_0 e^{-\theta\mu(20)}$.

Source: Aggarwal & Singh (3rd ed.), §7.3.2, Example 7.2, p. 336; companion slide 62.

Problem 7.7 · Logarithmic Model Execution Progress at 25 and 50 CPU Hours (Exercise 7.20)

Assume that the initial failure intensity is $\lambda_0 = 10 \text{ failures/CPU hr}$ and the failure intensity decay parameter is $\theta = 0.03/\text{failure}$. We have experienced 75 failures up to this time. Find the failures experienced and failure intensity after 25 and 50 CPU hours of execution.

1. **Given:**
   - Initial failure intensity: $\lambda_0 = 10 \text{ failures/CPU hr}$
   - Failure intensity decay parameter: $\theta = 0.03/\text{failure}$
   - Failures experienced at the stated current point: $\mu_P = 75 \text{ failures}$
   - Requested intervals: $\Delta\tau = 25$ and 50 CPU hours
2. **Required:**
   - Failures experienced and failure intensity at each requested point.
3. **Model & Assumptions:**
   - **Logarithmic Poisson Execution Time Model.** The main calculation treats 25 and 50 CPU hours as additional execution time from the stated current point. The alternate cumulative-time reading is also shown because the wording does not specify the reference point.
4. **Equations:**

$$\lambda_P = \lambda_0 e^{-\theta\mu_P}, \qquad \mu(\tau_P + \Delta\tau) = \mu_P + \frac{1}{\theta}\ln(1 + \theta\lambda_P\Delta\tau)$$

$$\lambda(\tau_P + \Delta\tau) = \frac{\lambda_P}{1 + \theta\lambda_P\Delta\tau}, \qquad \tau_P = \frac{e^{\theta\mu_P} - 1}{\lambda_0\theta}$$

   - If the requested times are cumulative from the start: $\mu(\tau) = \frac{1}{\theta}\ln(\lambda_0\theta\tau + 1)$ and $\lambda(\tau) = \frac{\lambda_0}{\lambda_0\theta\tau+1}$.
5. **Substitution & Working:**
   - Current state: $\lambda_P = 10e^{-0.03(75)} \approx 1.05399$ failures/CPU hr and $\tau_P = (e^{2.25} - 1)/0.30 \approx 28.29$ CPU hours.
   - **After an additional 25 CPU hours:** $\mu = 75 + \frac{1}{0.03}\ln(1 + 0.03(1.05399)(25)) \approx 94.42$ failures; $\lambda = \frac{1.05399}{1+0.03(1.05399)(25)} \approx 0.589$ failures/CPU hr.
   - **After an additional 50 CPU hours:** $\mu = 75 + \frac{1}{0.03}\ln(1 + 0.03(1.05399)(50)) \approx 106.61$ failures; $\lambda = \frac{1.05399}{1+0.03(1.05399)(50)} \approx 0.408$ failures/CPU hr.
   - **If 25 and 50 hours mean cumulative time from the start:** $\mu(25) \approx 71.34$, $\lambda(25) \approx 1.176$; $\mu(50) \approx 92.42$, $\lambda(50) = 0.625$. However, the stated 75 failures correspond to $\tau_P \approx 28.29$ hours, so the 25-hour point precedes the stated current state.
6. **Result:**
   - Main reading (25 and 50 additional CPU hours): $(\mu, \lambda) \approx (94.42, 0.589)$ and $(106.61, 0.408)$, respectively.
   - Cumulative-from-start reading: $(\mu(25), \lambda(25)) \approx (71.34, 1.176)$ and $(\mu(50), \lambda(50)) \approx (92.42, 0.625)$; this conflicts with the stated current point at 75 failures for $\tau = 25$.
7. **Check / Interpretation:**
   - Under the main reading, both requested times follow the present state and failure intensity decreases as execution time increases.

Source: Aggarwal & Singh (3rd ed.), Exercise 7.20, p. 363; companion slide 123.

Problem 7.8 · Comparative Calculation: Basic vs Logarithmic Poisson Model (Exercise 7.21)

The following parameters for the Basic Execution Time Model and the Logarithmic Poisson Execution Time Model are given:

- **Basic Model:** $\lambda_0 = 5$ failures/CPU hr, $v_0 = 125$ failures
- **Logarithmic Poisson Model:** $\lambda_0 = 25$ failures/CPU hr, $\theta = 0.30$/failure

Determine the additional failures ($\Delta\mu$) and additional execution time ($\Delta\tau$) required to reach the failure intensity objective of $\lambda_F = 0.10$ failure/CPU hr for both models (from their initial states λ_0).

1. **Given:**
   - Basic Model: $\lambda_{0,B} = 5$ failures/CPU hr, $v_0 = 125$ failures
   - Logarithmic Model: $\lambda_{0,L} = 25$ failures/CPU hr, $\theta = 0.30$/failure
   - Common target failure intensity: $\lambda_F = 0.10$ failure/CPU hr
2. **Required:**
   - For Basic Model: $\Delta\mu_B$ and $\Delta\tau_B$.
   - For Logarithmic Model: $\Delta\mu_L$ and $\Delta\tau_L$.
3. **Model & Assumptions:**
   - Comparison evaluated from initial state ($\lambda_P = \lambda_0$, $\mu_P = 0$) to target release objective (λ_F).
4. **Equations:**
   - **Basic Execution Time Model:**

$$\Delta\mu_B = \frac{v_0}{\lambda_{0,B}}(\lambda_{0,B} - \lambda_F) = v_0\left(1 - \frac{\lambda_F}{\lambda_{0,B}}\right)$$

$$\Delta\tau_B = \frac{v_0}{\lambda_{0,B}} \ln\left(\frac{\lambda_{0,B}}{\lambda_F}\right)$$

   - **Logarithmic Poisson Model:**

$$\Delta\mu_L = \frac{1}{\theta} \ln\left(\frac{\lambda_{0,L}}{\lambda_F}\right)$$

$$\Delta\tau_L = \frac{1}{\theta}\left(\frac{1}{\lambda_F} - \frac{1}{\lambda_{0,L}}\right)$$

5. **Substitution & Working:**
   - **Case 1: Basic Model Calculations:**
     - Scaling factor: $\frac{v_0}{\lambda_{0,B}} = \frac{125}{5} = 25$ CPU hrs.
     - Additional failures:

$$\Delta\mu_B = 125 \times \left(1 - \frac{0.10}{5}\right) = 125 \times (1 - 0.02) = 125 \times 0.98 = 122.5 \text{ failures}$$

     - **Additional execution time:**
       $\Delta\tau_B = 25 \times \ln\left(\frac{5}{0.10}\right) = 25 \times \ln(50) = 25 \times 3.912023 \approx 97.8006 \approx 97.80$ CPU hrs
   - **Case 2: Logarithmic Poisson Model Calculations:**
     - Scaling factor: $\frac{1}{\theta} = \frac{1}{0.30} = \frac{10}{3} \approx 3.3333$.
     - Ratio: $\frac{\lambda_{0,L}}{\lambda_F} = \frac{25}{0.10} = 250$.
     - Additional failures:

$$\Delta\mu_L = \frac{10}{3} \times \ln(250) = \frac{10}{3} \times 5.521461 = 18.40487 \approx 18.40 \text{ failures}$$

     - **Additional execution time:** $\Delta\tau_L = \frac{1}{0.30} \times \left(\frac{1}{0.10} - \frac{1}{25}\right) = \frac{10}{3} \times (10 - 0.04) = \frac{10}{3} \times 9.96 = 33.20$ CPU hrs
6. **Result:**
   - **Basic Model:** $\Delta\mu_B = 122.5$ failures, $\Delta\tau_B \approx 97.80$ CPU hrs.

- **Logarithmic Model:** $\Delta\mu_L \approx 18.40$ failures, $\Delta\tau_L = 33.20$ CPU hrs.

7. **Check / Interpretation:**
   - Although the Logarithmic Poisson model starts at a higher initial intensity ($\lambda_0 = 25$ vs 5), its high decay parameter ($\theta = 0.30/\text{failure}$) drops failure intensity far more rapidly per observed failure, requiring only ≈ 18.4 failures and 33.2 CPU hours compared to 122.5 failures and 97.8 CPU hours under the Basic model.

Source: Aggarwal & Singh (3rd ed.), Exercise 7.21, p. 364; companion slide 124.

Problem 7.9 · Calendar Time Component — Resource-Limited Execution Time Ratio (Examples 7.4–7.6)

During a software testing phase, resource usage is modeled as $\chi_k = \theta_k \tau + \mu_k \mu$. For the failure identification personnel resource ($k = I$):

- $\theta_I = 15$ person-hrs/CPU hr
- $\mu_I = 2$ person-hrs/failure
- Available personnel: $P_I = 2$ persons
- Utilization factor: $\rho_I = 0.80$ If an interval requires an additional execution time of $\Delta\tau = 10$ CPU hrs and experiences $\Delta\mu = 25$ failures, calculate:
- The total resource usage $\Delta\chi_I$ in person-hours.
- The calendar time Δt required to complete this testing phase (in calendar hours and 8-hour work days).

1. Given:

- Execution time increment: $\Delta\tau = 10$ CPU hrs
- Failure increment: $\Delta\mu = 25$ failures
- Effort parameters: $\theta_I = 15$ person-hrs/CPU hr, $\mu_I = 2$ person-hrs/failure
- Staffing and utilization: $P_I = 2$, $\rho_I = 0.80$

2. Required:

- Total resource requirement $\Delta\chi_I$ in person-hours.
- Calendar time Δt in calendar hours and working days.

3. Model & Assumptions:

- **Musa Calendar Time Component Model.**
- Testing is conducted in the Failure Identification Personnel Limited segment.
- 1 working day equals 8 productive calendar hours.

4. Equations:

$$\Delta\chi_I = \theta_I \Delta\tau + \mu_I \Delta\mu$$

$$\Delta t = \frac{\Delta\chi_I}{P_I \rho_I}$$

5. Substitution & Working:

- **Resource Usage:**

$$\Delta\chi_I = (15 \times 10) + (2 \times 25) = 150 + 50 = 200 \text{ person-hrs}$$

- **Calendar Time:**

$$\Delta t = \frac{200}{2 \times 0.80} = \frac{200}{1.60} = 125.0 \text{ calendar hrs}$$

- In 8-hour working days:

$$\Delta t_{\text{days}} = \frac{125}{8} = 15.625 \text{ work days} \approx 16 \text{ work days}$$

6. Result:

- $\Delta\chi_I = 200$ person-hrs
- $\Delta t = 125.0$ calendar hrs $(15.6$ working days$)$

7. Check / Interpretation:

- The interval-average ratio is $\Delta t/\Delta\tau = 125/10 = 12.5$ calendar hours per CPU hour. This agrees with average resource use of $200/10 = 20$ person-hours per CPU hour divided by available utilized capacity $P_I\rho_I = 1.6$ persons.

Source: Aggarwal & Singh (3rd ed.), §7.3.3, Examples 7.4–7.6, pp. 340–342; companion slides 72–82.

Problem 7.10 · Jelinski–Moranda Model & Bug Seeding Population Estimation (Example 7.7 & Exercise 7.26)

1. **Part A (Jelinski–Moranda Model):** A software system has an estimated initial fault count of $N = 50$, with a fault failure rate constant of $\phi = 0.002/\text{hr}$. Calculate the failure hazard rate $\lambda(t_i)$ when debugging starts ($i = 1$), after 10 faults have been removed ($i = 11$), and after 40 faults have been removed ($i = 41$).
2. **Part B (Harlan Mills Bug Seeding):** An organization seeds $S = 20$ artificial bugs into a software program before testing. Testing discovers a total of 30 bugs, of which $s = 12$ are seeded bugs and $n = 18$ are indigenous bugs. Estimate the total original indigenous bug population $\hat{N}$ and the number of unmanifested indigenous bugs remaining in the code.

Part A: Jelinski–Moranda Evaluation

1. **Given:**
   - Initial fault count: $N = 50$
   - Proportionality constant: $\phi = 0.002/\text{hr}$
   - Indices: $i = 1$ (first failure), $i = 11$ (11th failure / 10 removed), $i = 41$ (41st failure / 40 removed)
2. **Required:**
   - Hazard rate $\lambda(t_1)$, $\lambda(t_{11})$, and $\lambda(t_{41})$ in $\text{failures}/\text{hr}$.
3. **Model & Equation:**

$$\lambda(t_i) = \phi[N - (i-1)]$$

4. **Working:**
   - For $i = 1$:

$$\lambda(t_1) = 0.002 \times [50 - (1-1)] = 0.002 \times 50 = 0.10 \text{ failures}/\text{hr}$$

   - For $i = 11$:

$$\lambda(t_{11}) = 0.002 \times [50 - (11-1)] = 0.002 \times 40 = 0.08 \text{ failures}/\text{hr}$$

   - For $i = 41$:

$$\lambda(t_{41}) = 0.002 \times [50 - (41-1)] = 0.002 \times 10 = 0.02 \text{ failures}/\text{hr}$$

5. **Result (Part A):**
   - $\lambda(t_1) = 0.10 \text{ failures}/\text{hr}$
   - $\lambda(t_{11}) = 0.08 \text{ failures}/\text{hr}$
   - $\lambda(t_{41}) = 0.02 \text{ failures}/\text{hr}$

Part B: Bug Seeding Estimation

1. **Given:**
   - Seeded artificial bugs: $S = 20$
   - Discovered seeded bugs: $s = 12$
   - Discovered indigenous bugs: $n = 18$
2. **Required:**
   - Estimated total indigenous bugs $\hat{N}$.
   - Estimated remaining unmanifested indigenous bugs N_{rem}.
3. **Model & Assumptions:**
   - **Mills Bug Seeding Model (Capture–Recapture Analogy).**
   - Indigenous bugs and seeded bugs have identical detection probabilities and are randomly distributed throughout the code.
4. **Equation:**

$$\frac{s}{S} = \frac{n}{\hat{N}} \implies \hat{N} = \frac{S \times n}{s}$$

$$N_{\text{rem}} = \hat{N} - n$$

5. **Working:**

$$\hat{N} = \frac{20 \times 18}{12} = \frac{360}{12} = 30 \text{ indigenous bugs}$$

$$N_{\text{rem}} = 30 - 18 = 12 \text{ indigenous bugs remaining}$$

6. **Result (Part B):**
   - Estimated total indigenous bugs: $\hat{N} = 30$ bugs.
   - Estimated remaining indigenous bugs: $N_{\text{rem}} = 12$ bugs.
7. **Check / Interpretation:**
   - Recovery ratio for seeded bugs is $12/20 = 60\%$. Under equal detectability, the 18 discovered indigenous bugs also represent 60% of total indigenous bugs ($18/0.60 = 30$). Arithmetic is completely consistent.

Source: Aggarwal & Singh (3rd ed.), §7.3.4 & §7.3.5, Example 7.7, pp. 343–346; companion slides 89–93.
